

Research part for the proposal of an Scientific Event

Title: Biodiversity Dynamics in Coevolutionary Meta-ecosystems

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1. Scientific aspects

1.1 Introduction of the research subject

Coevolution, the adaptation resulting in reciprocal trait changes within and among species, has been invoked as a primary mechanism generating much of Earth's biodiversity (Althoff et al, 2014;Hembry et al. 2014). Two main theories have been proposed to govern the dynamics of coevolutionary diversification: Fisherian sexual selection theory where female and male traits can lead to sexual isolation and speciation, and the Geographic mosaic theory of coevolution which suggests that reciprocal changes of interaction traits among species produces coevolutionary diversification because interspecific interactions control the reproductive traits of the resource species. Despite the appealing of such theories for explaining much of Earth's biodiversity, the connections between coevolution of intra- and inter-specific interactions across heterogeneous landscapes to map biodiversity dynamics in species- and interaction-rich ecological networks remain elusive (Thompson 2014).

There is a critical need to join theoretical and empirical evolutionary ecologists to take into account the empirical evidence of reciprocal trait changes for the stability and of species-rich communities. The aim of the workshop is to integrate species spatial interaction data to new process-based coevolution models with different strengths of reciprocal trait change within and between species. This integration would allow to explore the patterns of coexistence of co-evolved species, their role in ecosystem function, and the risk of collapse of co-evolved ecosystems. This is a critical need for four reasons. First, the extinction rate consequence of human-disturbances is at an unprecedented rate, yet, most models guiding the predictions of extinction rates often assume that species evolve independently of one another (Thompson 2014). Second, despite eco-evolutionary studies of species interactions are increasingly common, it is difficult to connect co-evolving trait distribution within and between species at small scales to the large-scale associations between

coevolutionary hot spots, biodiversity function and meta-ecosystem dynamics, and third, although many experiments and empirical studies have shown convergence of time scales between ecological and coevolutionary processes (Hairston et al. 2005, Fussmann et al. 2007, Schoener 2011; Thompson 2013), there is a gap in understanding how ecological and coevolutionary dynamics in species-rich assemblages shape biodiversity and ecosystem function in a spatio-temporal context – the meta-ecosystem.

A meta-ecosystem is a set of local ecosystems connected by spatial flows of energy, materials and organisms across ecosystem boundaries (Loreau et al. 2003). One of the main challenges in theoretical and applied Ecology is to incorporate the paradigm of meta-ecosystem ecology to improve our understanding of the coevolutionary processes within and between species and their role in ecosystem functioning. The meta-ecosystem concept provides a powerful tool to understand the properties that arise from spatial coupling of local ecosystems, such as coevolutionary selection mosaics, stabilization of ecosystem processes and indirect interactions at landscape or regional scales (Gravel et al 2010). We propose a coupled workshop and research visit aiming to integrate empirical datasets, theory and analytical methods to bridge the gap between coevolutionary and ecological spatio-temporal dynamics of species-rich interaction networks, which will help us to advance our understanding of ecosystem functioning across spatial and temporal scales.

The timing for such a workshop has never been better. There has recently been tremendous advancement in mathematical approaches to studying coevolution in interacting species-rich communities (Althoff et al, 2014, Hembry et al. 2014). Advances include better methods for studying trait evolution, species interaction networks, and the relationship between the two (de Andreazzi et al., 2018). However, most of these methods are, at best, indirectly connected to biodiversity dynamics and function in meta-ecosystems. One reason for this is that current approaches almost all assume that species coevolve in isolated communities. This assumption, more for mathematical convenience than biological accuracy, is limiting the field to study relationships among coevolution, biodiversity dynamics and ecosystem function across spatial-temporal scales in species-rich interacting systems

We will combine various methods and models to merge co-evolving trait distributions and species interactions in meta-ecosystems (de Andreazzi et al., 2018, Melián et al., 2018). Such approaches will consider spatially variable biotic selection and individuals migration among heterogeneous patches to generate complex selection mosaics, leading to spatial variation in species traits, and

therefore to predictions of the structure of ecological networks in meta-ecosystems (Gravel et al. 2016, Massol et al. 2017). For this workshop, we will seek to push the field further, challenging the participants to synthesize spatial and trait-interaction data in species-rich ecosystems to test the methods with empirical data to investigate coevolutionary dynamics and biodiversity function at different spatial scales.

1.2 Scientific aims and methods of the workshop

We will focus on implementing new analysis tools and applying them to the integrated datasets. We propose bringing together biologists skilled in mathematics, software development, and researchers with expertise in model systems and field data. We focus on the following three questions in co-evolution in meta-ecosystems.

Question 1: How does reciprocal trait change influence biodiversity dynamics and ecosystem function in species-rich interacting meta-ecosystems?

We aim to integrate the effects of ecological, coevolutionary, and migration dynamics in shaping biodiversity in meta-ecosystems. We will merge the meta-ecosystem perspective (Gravel et al. 2016, Massol et al. 2017, Melián et al. 2018) into existing models of coevolutionary interaction networks (Guimarães et al. 2011, Andreazzi et al. 2017). We will run simulations to predict biodiversity distribution across space and time under different scenarios assuming weak and strong reciprocal trait changes (*i.e.*, *trait matching and exploitation barriers*). These scenarios would allow us to link the strength of coevolution between species to ecosystem function like productivity and energy flows. Metacommunity and meta-ecosystem theory approaches will complement previous scenarios to understand how spatial processes governed by dispersal (*i.e.* mass effect) or niche width (*i.e.* species sorting, patch dynamics) can alter the coevolutionary dynamics within each community by contributing to species trait variation and network structure and stability, and thus meta-ecosystem function (Mouquet and Loreau 2003, Leibold et al. 2004, Logue et al. 2011, Winegardner et al. 2012, Borthagaray et al. 2014 a,b, 2015a, Astegiano et al. 2015a, Melian et al. 2015b, Laroche et al. 2016, Massol et al. 2017 a, b).

Question 2: How do spatial connectivity/arrangement of ecosystems and coevolutionary dynamics interact to facilitate rapid biodiversity changes of in meta-ecosystems?

Most interactions between species generate reciprocal selection pressures, and the coevolutionary dynamics that emerges over time varies with the nature of the ecological interaction (i.e., competitive, mutualistic, or antagonistic), the architecture of the coevolving interactions between species (Wade 2007, Loeuille and Leibold 2008, Doebeli 2011, Nuismer et al. 2013, Althoff et al. 2014, Hembry et al. 2014, Allhoff and Drossel 2016), and the spatial connectivity of the landscape. Yet, the connection between traits and coevolution at local scales and the connectivity of the landscape for produce rapid biodiversity changes and coevolutionary hot spots are not well understood (Wade 2007, Doebeli 2011, Althoff et al. 2014, Hembry et al., 2014). We will run simulations under landscapes with different connectivities/arrangements and migration scenarios to disentangle the interplay between coevolutionary dynamics at the local level and the rapid changes of biodiversity at the landscape level.

1.3 Expected results

We will produce an opinion, review or ideas-perspectives paper entitled “*Biodiversity dynamics in coevolutionary meta-ecosystems*” for submission to *Trends in Ecology and Evolution*, *Ecology Letters* or *Biological reviews*. The two questions above will form the core of the discussions at the workshop, and they will help drafting an outline for the paper during the meeting. Prior to the meeting, attendees will provide written perspectives about how to address each of the overarching questions and the type of spatial trait-interaction-data they can provide. On the first day, participants will present these ideas to a broader audience based on their own research programs, and then on subsequent days we will synthesize these perspectives into a single review/opinion and synthesis style paper integrating the datasets to the methods to be implemented during the workshop. Overall, the workshop will provide the seeds of many future collaborations among attendees, which have been carefully chosen to not have gender bias (6F:6M, from 10 countries), represent diverse fields (e.g. theoretical ecologists, evolutionary biologists, and community ecologists) and scientific career stages (2 Prof., 4 Associate Prof. or group leaders, 2 research scientists, 3

postdoctoral researches, and 1 PhD student).

2. Organisational aspects

2.1 Milestones and program (Table 1)

The main milestones of the workshop will be:

1. Draft the outline for a synthesis paper about Biodiversity dynamics in coevolutionary meta-ecosystems summarizing the main findings obtained during the workshop.
2. Discuss future directions for the field that unify short- and long-term approaches integrating modeling and data
3. Develop new software approaches and make them available throughout github and related repositories to analyse data
4. Develop new ideas to strengthen future grant proposals and collaborations

2.2 Venue

The workshop will be hosted at Eawag's Center for Ecology, Evolution & Biogeochemistry (CEEB), in Kastanienbaum. This is an ideal venue. All the invited guests can stay on-site in Eawag apartments, directly on the shore of Lake Lucerne. The meeting facilities are very suitable and have been used before for such workshops. We have previously used this location to host several small workshops and student courses with great success. Such events also help to raise our international profile, and, in the past, have led to repeat visits from graduate students and visiting faculty on sabbatical terms.

2.3 Other funding sources

Eawag will provide the necessary infrastructure for the workshop (venue, consumables, IT, etc...). No other funds are needed to support this workshop.

3. Partnership aspects

3.1 Complementarity aspects

We have selected a list of participants that represent a wide range of ages, expertise, and levels of experience. We have identified a mixture of people with (i) skills in mathematical biology, (ii) experience with software development, and (iii) strong conceptual skills in evolutionary biology, model systems and field studies.

3.2 Potential for further collaboration

We expect that this workshop will lead to new international collaborations. In particular, we have found in our own work that coevolution in antagonistic networks might have different effects on meta-ecosystem dynamics and function,. We see great potential in linking the research of the participants together specifically, the interplay between coevolution in local ecosystems with the landscape dynamics and connectivity. Furthermore, we are in the middle of a great revolution in our understanding of coevolution among many interacting species at different spatial scales and its effects on biodiversity dynamics and ecosystem function. Yet, the integration between robust empirical patterns and process-based models, the main aim of this workshop, is missing.

3.3 References

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